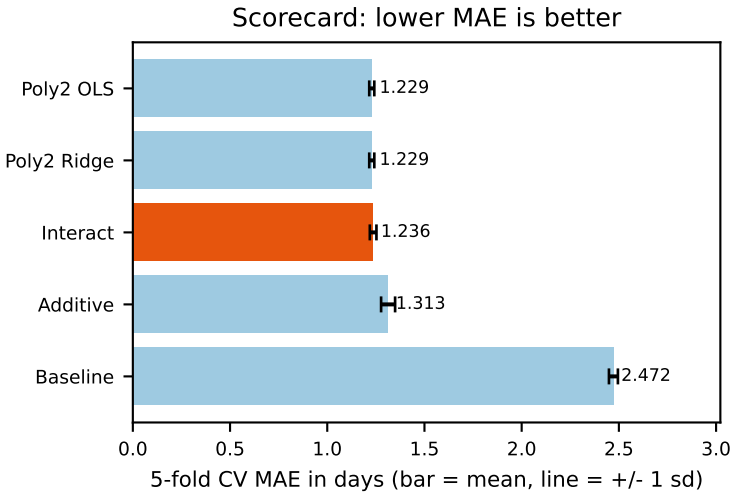
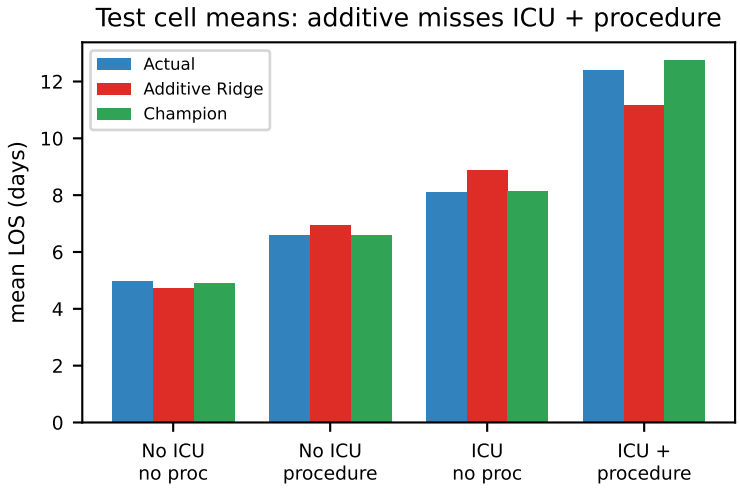
Day 24 Cheat Sheet: Interactions and Polynomials

Week 4 Day 3: when ICU plus a procedure is more than the sum of the parts

TERM	WHAT IT MEANS	CODE
Interaction	extra effect when two features are both on	icu_stay * procedure
Quadratic / polynomial	a curved effect (age bends, not a straight line)	age ** 2
PolynomialFeatures	builds all products (and squares) up to a degree	PolynomialFeatures(degree=2)
interaction_only=True	pairwise products only; no x_i^2 terms	interaction_only=True
Full degree 2	products AND squares (8 cols -> 44 terms here)	interaction_only=False
Scale AFTER expanding	age^2 and 0/1 products live on different scales	then StandardScaler()
Ridge after expanding	extra terms fit noise unless you penalize them	RidgeCV(...)
Interaction leftover	(1,1) - (1,0) - (0,1) + (0,0) from a 2x2 table	~2.81 days here
Simplicity rule	lowest CV MAE; prefer simpler within 1 fold sd	write BEFORE opening test





THIS DATA: 4,000 simulated discharges (seed 24), 8 features, continuous los_days, 75/25 split
Train LOS mean 6.12 days (sd 3.16). Raw 8 -> 36 interaction terms -> 44 full degree-2 terms.
ICU x procedure cell means (all rows): no/no 4.82 | no/proc 6.51 | ICU/no 8.11 | both 12.61
Interaction leftover 2.81 days. Share ICU 18.6%, procedure 34.0%, both 6.4%.
5-fold CV MAE: Additive Ridge 1.313 | Interactions 1.236 | Poly2 Ridge 1.229 | baseline 2.472
Alphas: Additive 20.31 | Interactions 0.001 | Poly2 0.838
Champion: Interactions + Ridge (simplicity rule). Test once: MAE 1.249 days, RMSE 1.555, R2 0.761
Within 2 days: 80%. Median |error| 1.06 days. ICU+procedure test n=71: additive MAE 1.833 vs champion 1.350.

HABITS THAT KEEP YOU HONEST
- Plot the 2x2 cell means first. If (ICU + procedure) is not the sum of the parts, an additive model will miss a subgroup.
- Expand with PolynomialFeatures, THEN scale, THEN regularize (Day 23). Eight columns become 36 or 44 very fast.
- Write the selection rule before test: lowest CV MAE, prefer the simpler expansion within 1 fold sd.
- Report overall MAE and the MAE on the interaction cell. A small average gain can hide a large gain for the right patients.
- Read recovered terms (procedure icu_stay, acuity icu_stay). Correlated extras like age*comorbidities can stand in for a curve.

```
pipe = make_pipeline(
    PolynomialFeatures(degree=2, include_bias=False, interaction_only=True),
    StandardScaler(),
    RidgeCV(alphas=np.logspace(-3, 3, 40), scoring="neg_mean_absolute_error", cv=cv))
```

Infographic: Extra days hiding in one cell

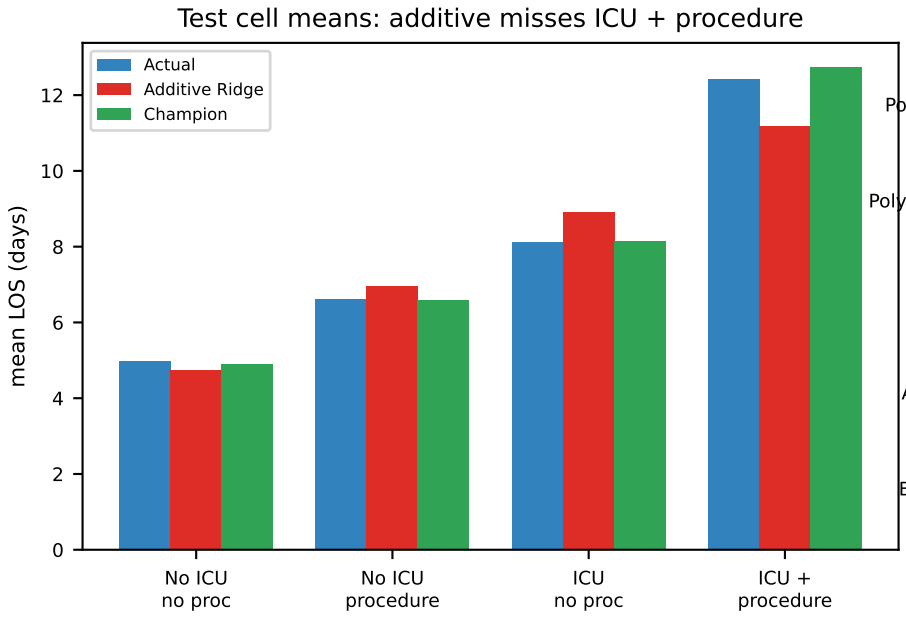
Additive models average the ICU-plus-procedure bump across everyone

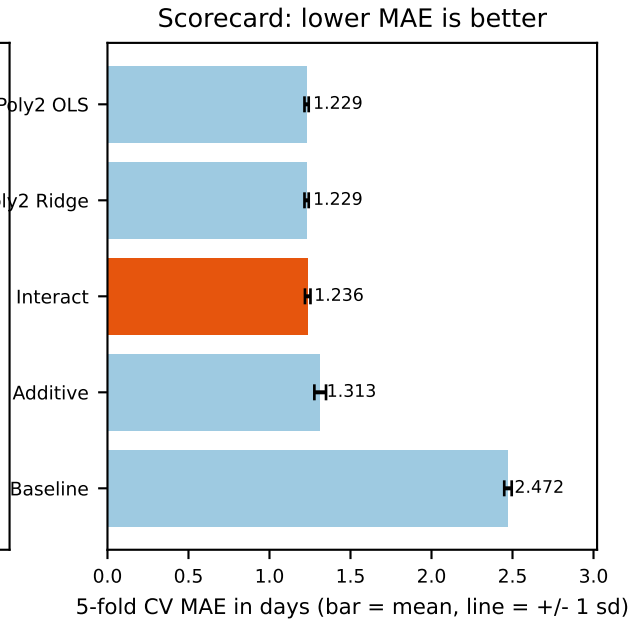
LEFTOVER
2.81 d

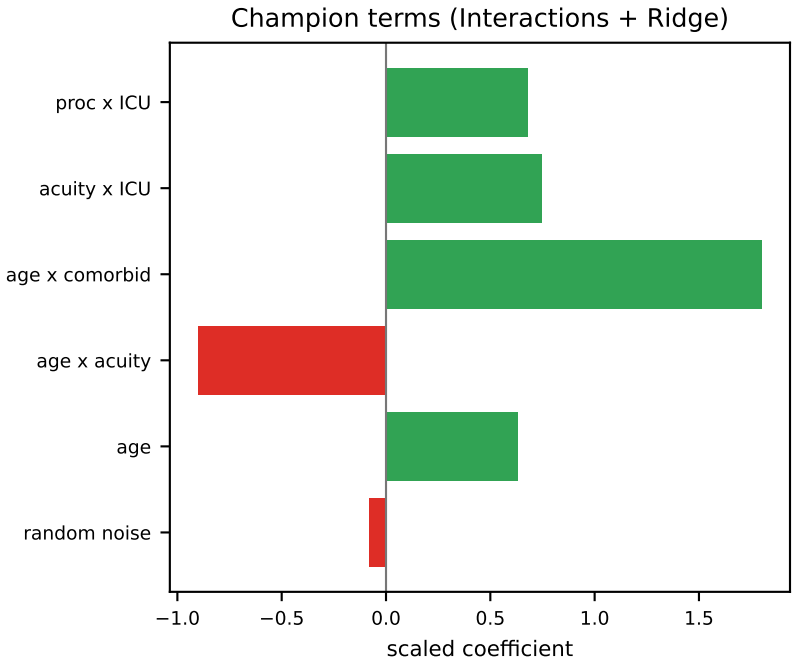
CHAMPION
Interact

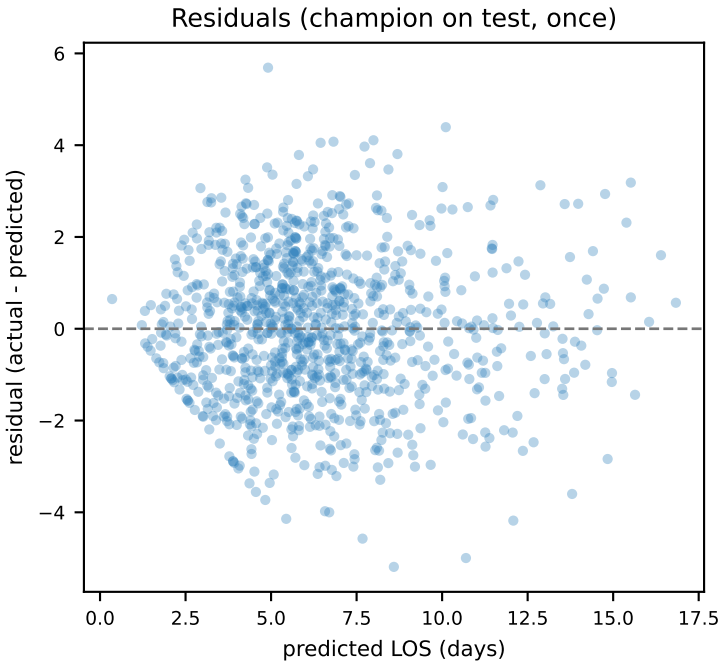
TEST MAE
1.25 d

ICU+PROC
1.35 d









Takeaways
- Cell means are not additive: both-together is 12.61 days, leftover 2.81 after subtracting the two main effects.
- Additive Ridge predicts about 11.18 days for ICU + procedure on test (actual 12.41). The champion recovers 12.74.
- Overall test MAE only moves from 1.335 to 1.249, but on the 71 ICU-plus-procedure test patients it drops from 1.833 to 1.350.
- Interactions + Ridge wins the simplicity rule over Poly2 (1.236 vs 1.229, within 1 fold sd). Extra squares were not worth it today.
- Tomorrow: learning curves and validation curves, to see whether extra polynomial terms fit signal or memorize the training set.

Hands-On Worksheet: ds-genai-90-days / day24

60-90 minutes | notebook: day24/interactions_los.ipynb + day24/README.md

[ ] mkdir day24, conda activate ds90, open day24/interactions_los.ipynb

[ ] Simulate ~4,000 discharges with true interactions and age^2 (snippet 1, seed 24)

[ ] Print the ICU x procedure cell means and the interaction leftover

[ ] Define models: baseline, additive Ridge, interactions+Ridge, poly2+Ridge, poly2+OLS

[ ] Confirm term counts: 8 raw, 36 interaction-only, 44 full degree-2

[ ] Run 5-fold CV MAE/RMSE/R2 on TRAIN (snippet 3) and record alphas

[ ] Write the simplicity rule BEFORE opening test (prefer simpler within 1 sd)

[ ] Pick champion, score test once, print MAE/RMSE/R2 (snippet 4)

[ ] Read recovered terms: procedure icu_stay and acuity icu_stay should rank high

[ ] Compare test cell means: actual vs additive vs champion (snippet 5)

[ ] Compute MAE on the ICU+procedure test rows (not just overall MAE)

[ ] README: leftover, scorecard, cell-mean chart, champion and why, limits

[ ] git add day24 && git commit && git push (snippet 6)

[ ] Stretch: Snowflake AVG/ABS error grouped by ICU and procedure (snippet 7)

```
pipe = make_pipeline(
    PolynomialFeatures(degree=2, include_bias=False, interaction_only=True),
    StandardScaler(),
    RidgeCV(alphas=np.logspace(-3, 3, 40), scoring="neg_mean_absolute_error", cv=cv))
```

CV MAE Additive: _____ Interact: _____

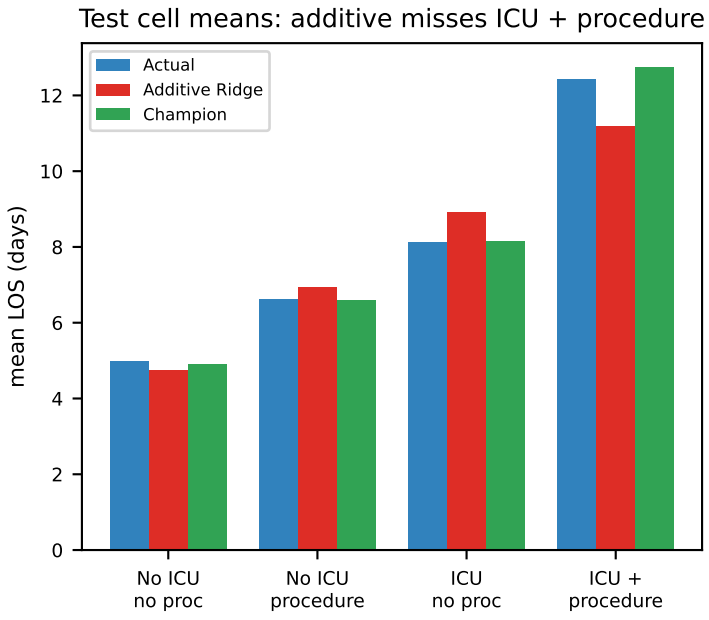
Poly2: _____ baseline: _____

Champion: _____ Test MAE: _____ d

RMSE: _____ R2: _____

Leftover: _____ d

ICU+proc MAE additive _____ champion _____



Day 24 Self-Check Quiz

1. What does icu_stay * procedure mean in a length-of-stay model?

_____

2. How can overall MAE look only a little better while one patient group improves a lot?

_____

3. What does PolynomialFeatures(degree=2) add beyond the original columns?

_____

4. What is the difference between interaction_only=True and False?

_____

5. Why scale AFTER expanding, not before?

_____

6. Why regularize after a polynomial expansion (hint: Day 23)?

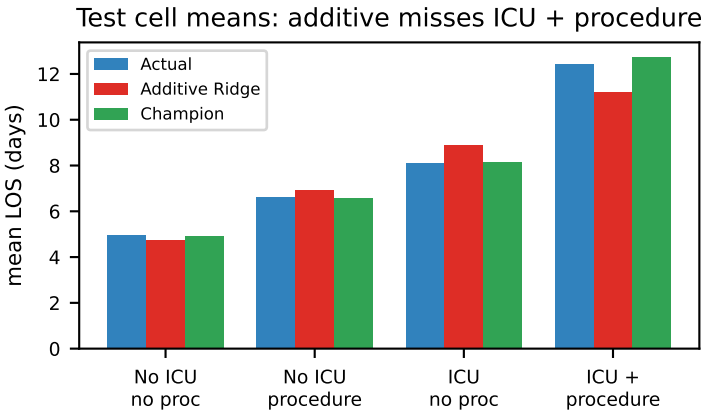
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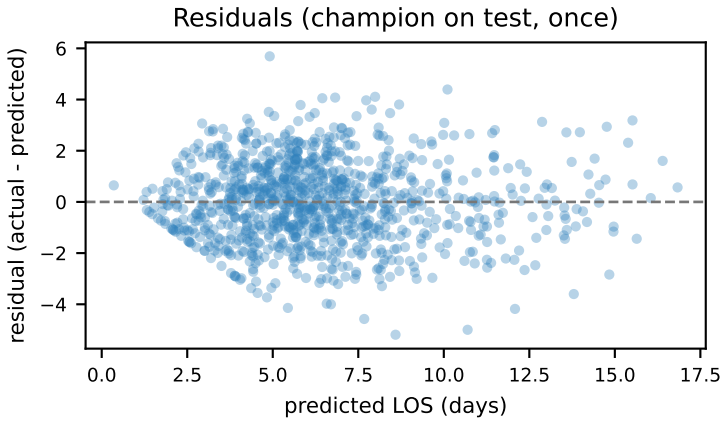
7. What simplicity rule picked the champion today?

_____

8. How do you compute the interaction leftover from a 2x2 table of group means?

_____





Answers:
1 extra days (or fewer) when both happen, beyond adding the two main effects
2 overall MAE averages many ordinary stays; the extra days sit in a smaller ICU-plus-procedure cell
3 all pairwise products x_i*x_j and, unless interaction_only, the squares x_i^2
4 True keeps products only (36 terms here); False also adds squares (44 terms here)
5 expanded columns have very different raw scales (age^2 vs a 0/1 product); Ridge assumes comparable scale
6 8 columns become 36 or 44 terms that can fit noise unless you penalize them
7 lowest CV MAE, then prefer the simpler expansion if it is within 1 fold sd of the best
8 (ICU1,Proc1) - (ICU1,Proc0) - (ICU0,Proc1) + (ICU0,Proc0); about 2.81 days on this data